Storing Big Data

In this lesson, we will go over different ways to store big data and how the cloud makes this easier. From traditional databases to object storage, you'll learn how systems like AWS S3, Azure Blob Storage, and GCP Cloud Storage are used in real-world big data platforms. We'll also touch on data lakes and how to structure them.

Learning Objectives

- Differentiate between file, block, and object storage systems.
- Understand the concept and benefits of data lakes.
- Describe how big data is ingested, stored, processed, and consumed in cloud platforms.
- Explore the concept of schema-on-read and medallion architecture in cloud data lakes.
- Understand governance and security best practices for data stored in the cloud.

Storage Systems: File Vs. Block Vs. Object

Data storage is fundamental to any computing system. Before cloud-native data lakes existed, data was stored in either files or blocks on physical servers and disks. Nowadays, cloud providers give us access to scalable object storage with zero upfront investment—saving us from the complexities of managing and maintaining infrastructure.

Let's compare the key differences:

Storage Type	Structure	Best For	Examples
File	Hierarchy (folders/files)	Simple document storage	NFS, SMB
Block	Fixed-size chunks (blocks)	Databases, high performance	HDD/SSD, SAN
Object	Data + metadata	Unstructured big data	AWS S3, Azure Blob

Data Lakes and Cloud Storage



Figure 1: Let's talk about lakes

A data lake is a cloud-native architecture for storing structured, semi-structured, and unstructured data in its raw form. Unlike a data warehouse, which requires a fixed schema, a data lake applies schema only when data is read or queried—this is known as schema-on-read.

Cloud data lakes typically separate compute from storage, allowing us to scale each independently. The three main storage zones are:

- **Bronze:** Raw data—ingested as-is, may be messy or redundant.
- **Silver:** Cleaned and filtered data, often enriched and transformed.
- **Gold:** Trusted business-level datasets ready for analytics or ML.

Popular cloud storage services that support data lake architectures include:

- **AWS:** S3 for storage, Glue for transformation, Athena for querying
- **Azure:** Blob Storage with Synapse or Databricks
- **GCP:** Cloud Storage with BigQuery or Dataflow

Ingestion, Processing, and Consumption

Storing data is only the beginning. You need ways to get data into the lake (ingestion), prepare it (processing), and make it useful (consumption).

Ingestion

- Batch tools like Apache NiFi, Azure Data Factory, or AWS Glue ingest files, logs, and DB extracts.
- Streaming platforms like Kafka, Spark Streaming, or Kinesis are used for real-time ingestion.

Processing

- Transformation tools (Spark, dbt, Glue) convert raw data into structured, usable formats (e.g. Parquet).
- Bronze → Silver → Gold transformation logic may include deduplication, validation, joins, aggregations.

Consumption

- Business Intelligence (BI) tools like Power BI, Tableau, or Looker query gold-layer data.
- ML tools or notebooks (SageMaker, Databricks) use gold/silver layers for model building.

Governance and Security

Security and compliance are non-negotiables for big data storage.

- **IAM:** Enforce access via roles and policies (e.g. AWS IAM, Azure AD).
- **Encryption:** Encrypt data at rest and in transit (e.g. KMS, SSE).
- **Monitoring:** Log access and activity using CloudTrail, GuardDuty, Azure Monitor.
- **Firewall & Threat Protection:** Use tools like AWS WAF to protect applications against the most common web exploits and threats.

Conclusion

Storing big data is not just about saving files—it's about enabling analytics at scale. By understanding object storage, schema-on-read, ingestion tools, and the medallion architecture, you can build scalable, cost-effective pipelines in the cloud.

Key Takeaways

- **Takeaway 1:** Object storage like S3 and Blob is ideal for unstructured big data.
- **Takeaway 2:** Data lakes support schema-on-read and allow flexible ingestion.
- **Takeaway 3:** Medallion architecture helps clean and organize lake data in zones (Bronze, Silver, Gold).
- **Takeaway 4:** Tools like Spark, Glue, and BigQuery power processing and analytics.
- **Takeaway 5:** Governance, encryption, and access control are vital for cloud-scale storage.